

- 3** A bob of mass 100 g is tied to a brass weight of mass 400 g using an inelastic string passed through a smooth vertical tube. The bob is made to move in a circular motion in the horizontal plane as shown in Fig. 3.1, such that the part of the string from the top of the tube to the bob is kept at 30 cm long and is at an angle θ to the vertical.

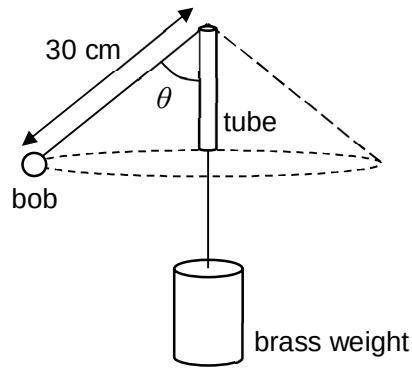


Fig. 3.1

- (a)** The bob moves at a constant speed v .

Explain why it has an acceleration.

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..... [2]

- (b)** Calculate θ . Explain your working.

$\theta = \dots\dots\dots^\circ$ [3]

(c) Determine v .

$v = \dots\dots\dots \text{ m s}^{-1}$ [2]

4 (a) Explain what is meant by the *internal energy* of a gas